

Omise: Providing easy- to-deploy payment gateway services for merchants

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Google Cloud results

- Achieved availability of 99.99%
- Saved 10 hours a week in sys
- Reduced network latency from 200 to 250 milliseconds

Founded in 2014 as an e-commerce company, Omise has evolved to provide payment gateway services for businesses in Southeast Asia. The platform includes the tools needed to convert sales into payments from customers. Headquartered in Bangkok, Thailand and operating out of hubs in Japan, and Singapore, Omise has



About Omise

Omise is an online payments solution platform that allows businesses to securely accept payments over the internet. Omise currently operates in Thailand and Japan, and the service will soon be available in Indonesia, Singapore, and Malaysia.

Industries: Financial Services & Insurance

to 90 employees and supports thousands of merchant accounts. Location: Thailand

"We developed our gateway in the first half of July 2014, and complied with the Thai Industry Data Security Standard to provide payment services," explained Alexandre Frederico Araujo, Chief Information Officer, "and we successfully processed our first credit card transaction in November 2014 and subsequently received a license from the Bank of Thailand that allowed us to operate a full-scale service." The Omise customer base comprises businesses of all sizes that appreciate the ease with which they can access tools to customize checkout forms without compromising security, integrate payment forms, and enable one-click payments.

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)



Challenge

To enable businesses to sell products and services online, Omise set down key criteria for its underlying technology. These criteria included near 100 percent uptime and enabling the rapid processing and completion of payments. Omise started operations in a public cloud service but found its costs

escalated quickly. Following a review, Omise decided to move its payment gateway environment to Google Cloud. "We opted for Google Cloud for three key reasons: we only had to use what we needed when we needed to, the performance met our requirements, and we could scale easily," says Araujo.

Solution

Omise was most interested in running Docker containers on Container Engine, developed using Kubernetes. "We determined that this platform would allow us to complete deployments quickly and often and eliminate concerns about the underlying infrastructure," says Araujo. The business migrated its payment gateway and associated systems to Google Cloud in the Asia Pacific Asia-East 1 Region in Taiwan. Omise is now running a Container Engine cluster comprising Compute Engine instances running Kubernetes. Eighty percent of the cluster is dedicated to production and the remaining 20 percent is used for testing and staging. Node instances managed from a master endpoint provide the services for the Docker containers used to run the Omise applications and associated systems.

Running its microservices-based applications in Kubernetes has enabled Omise to cut the time for developers to release new versions of their software from up to two hours to a few minutes. The business uses a Jenkins automation server to enable continuous integration and deployment to

Kubernetes and has gained the ability to monitor the number of requests directed to individual Docker containers.

Reduced network latency and high availability delivered through Google Cloud

Moving to Container Engine has enabled Omise to reduce network latency from 400 milliseconds to only 250 milliseconds, increasing the speed with which payments are processed through its gateway services and improving the experience for merchants and consumers. Furthermore, Omise has been able to maintain availability levels at 99.999 percent to ensure that the payment services critical to the viability of most businesses selling online can be accessed as needed. The business has also been able to scale quickly to support spikes in demand that occur when customers undertake promotions or make available tickets to high-demand events.

"In some cases, we have had to support up to 3,000 credit card transactions in a minute when tickets are made available to in-demand artists," says Araujo.

"We have been able to do this by creating preview Docker images of our architecture that can be deployed in seconds; in fact we can double our architecture in just five minutes if necessary."



Omise has also been able to use Google Cloud firewalls to control access to its Compute Engine instances. This has helped protect highly sensitive merchant and payment data from leakage or unauthorized access. "Being able to secure this data is critical to our reputation with merchants and their customers as a safe, reliable provider of payment services," says Araujo. "Thanks to Google Cloud, the data is in safe hands."

While the business does not directly attribute revenue gains to running its payment gateway service on Google Cloud, Araujo points to the stability and performance of the platform as a key contributor to improved sales. He is more definitive about the reductions in system administration achieved since moving to Google Cloud: "I estimate we've saved about 10 hours a week that we are now devoting to building our core business," he says. Overall, the business is extremely pleased with Google Cloud and believes the culture underpinning the cloud platform aligns closely with its own direct, collaborative approach.